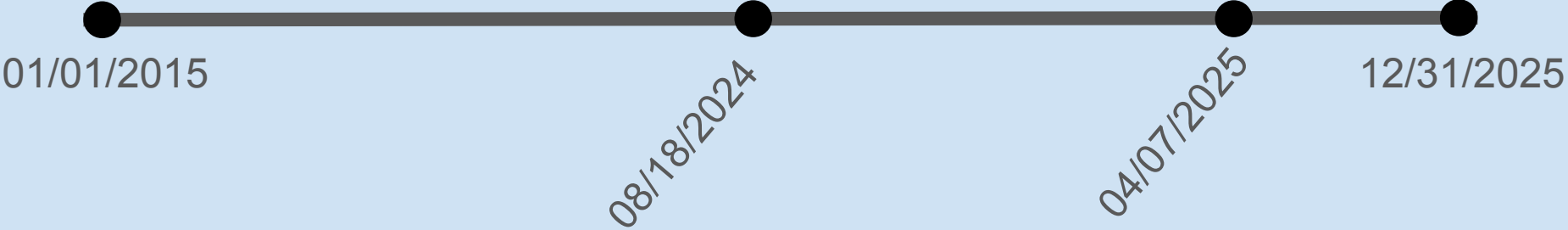


From 2015 to 2025:
Evaluating Changes in
Public Safety Call
Outcomes in Eugene

By: Brooke Watson

Background



Background

MCSLC



01/01/2015

08/18/2024

04/07/2025

12/31/2025

Background

MCSLC



01/01/2015

08/18/2024

04/07/2025

12/31/2025

RQ: How do call outcomes differ in Eugene, relative to Springfield, after the implementation of MCSLC and end of CAHOOTS in Eugene?

Eugene Data

yr	agency	service	inci_id	calltime	case_id	callsource	nature	closecode	closed_as
2017	EPD	LAW	17000001	2017-01-01 0:00:16	NULL	E911	SHOTS FIRED	REL	REFERRED TO OTH
2017	EPD	LAW	17000002	2017-01-01 0:01:26	NULL	PHONE	ILLEGAL FIREWORK	QOA	QUIET ON ARRIVAL
2017	EPD	LAW	17000003	2017-01-01 0:01:43	NULL	SELF	TRAFFIC STOP	ADVI	ADVISED
2017	EPD	LAW	17000005	2017-01-01 0:02:17	NULL	E911	ILLEGAL FIREWORK	INFO	INFORMATION ONL
2017	EPD	LAW	17000008	2017-01-01 0:02:45	1700004	W911	DISPUTE	ARR	ARREST
2017	EPD	LAW	17000009	2017-01-01 0:02:53	NULL	SELF	TRAFFIC STOP	SBCK	SOBRIETY CHECK
2017	EPD	LAW	17000011	2017-01-01 0:03:12	NULL	SELF	TRAFFIC STOP	UTC	UNIFORM TRAFFIC

Eugene Data

yr	agency	service	inci_id	calltime	case_id	callsource	nature	closecode	closed_as
2017	EPD	LAW	17000001	2017-01-01 0:00:16	NULL	E911	SHOTS FIRED	REL	REFERRED TO OTH
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2017	EPD	LAW	17000005	2017-01-01 0:02:17	NULL	E911	ILLEGAL FIREWORK	INFO	INFORMATION ONL
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2017	EPD	LAW	17000011	2017-01-01 0:03:12	NULL	SELF	TRAFFIC STOP	UTC	UNIFORM TRAFFIC



ID	City	Dispatch Date & Time	closed_as	calltime	agency
12236.0	Eugene	2025-05-12 13:34:00	ARREST	2025-05-12 13:34:00	MCSLC
12888.0	Eugene	2025-07-25 21:14:00	ARREST	2025-07-25 21:14:00	MCSLC
10504.0	Eugene	2025-03-29 22:43:00	ARREST	2025-03-29 22:43:00	MCSLC
12220.0	Eugene	2025-04-26 18:54:00	ARREST	2025-04-26 18:54:00	MCSLC
12224.0	Eugene	2025-05-12 20:51:00	ARREST	2025-05-12 20:51:00	MCSLC

Eugene Data Final

calltime	closed_as	agency	primeunit	is_cahoots
2015-01-01 00:00:00	ASSISTED	EPD	_5E48	False
2015-01-01 00:00:44	RESOLVED	EPD	_3F65	False
2015-01-01 00:01:05	ASSISTED	CAHE	_3J79	True
2015-01-01 00:03:16	PATROL CHECK	EPD	_5E48	False
2015-01-01 00:03:34	ADVISED	EPD	_5K97	False

Springfield Data

A	B	C	D	E	F
Incident Number	Initial Call Type	Final Call Type	Responding Agency	Primary Responding Unit	Call Creation Time
20000001	ILGLFR	ILLEGAL FIREWORKS	SPD		01:22.0
20000002	TRFSTP	DWS	SPD	1S18	01:43.0
20000005	TRFSTP	TRAFFIC STOP	SPD	1S23	03:12.0
20000008	SHTFIR	SHOTS FIRED	SPD	1S13	05:22.0
20000014	TRFSTP	TRAFFIC STOP	SPD	3D1	10:28.0
20000015	TRFSTP	TRAFFIC STOP	SPD	1S22	10:50.0
20000017	ILGLFR	ILLEGAL FIREWORKS	SPD	1S18	12:33.0
20000022	HARAS	HARASSMENT	SPD	1S13	15:42.0

Close Code	Definition
2012	Executive Order 2012 Violation
DIS	Disregard
ABAN	Abandoned Vehicle
ACTT	Action Taken
ACC	Accidentally Chose New Event
ADVI	Advised
AID	Medical Aid EPD
AO	Assisting Officer
ARR	Arrest
ARU	ARU Contacted and Will Follow Up
ASST	Assisted
AWO	Animal Welfare Officer

A	B	C	D	E	F	G	H
inci id		call time	prime unit	units dispatched in order			
15000004		01/01/2015 00:01	1S12	1S12			
15000013		01/01/2015 00:04					
15000022		01/01/2015 00:11	3S12	4S73 ,3S12 ,3S18			
15000026		01/01/2015 00:14	1S12	1S12			

ID	City	Dispatch Date & Time	closed_as	calltime	agency
14472.0	Springfield	2025-11-04 05:42:00	EMERGENCY DEPARTMENT	2025-11-04 05:42:00	MCSLC
12245.0	Springfield	2024-08-22 00:00:00	EMERGENCY DEPARTMENT	2024-08-22 00:00:00	MCSLC
10052.0	Springfield	2025-01-14 15:04:00	EMERGENCY DEPARTMENT	2025-01-14 15:04:00	MCSLC
10062.0	Springfield	2025-01-16 15:19:00	EMERGENCY DEPARTMENT	2025-01-16 15:19:00	MCSLC
10070.0	Springfield	2025-02-01 16:31:00	EMERGENCY DEPARTMENT	2025-02-01 16:31:00	MCSLC

Springfield Data Final

agency	prime unit	Call Year	closed_as	calltime	yr	month	is_cahoots
SPD	1S12	2015.0	ASSISTED	2015-01-01 00:01:00	2015.0	1.0	False
SPD		2015.0	DISREGARD	2015-01-01 00:04:00	2015.0	1.0	False
SPD	3S12	2015.0	ARREST	2015-01-01 00:11:00	2015.0	1.0	False
SPD	1S12	2015.0	TRAFFIC STOP ONLY	2015-01-01 00:14:00	2015.0	1.0	False
SPD	1S18	2015.0	TRAFFIC STOP ONLY	2015-01-01 00:17:00	2015.0	1.0	False

Methods

1

Obtain and clean
data

2

Are call outcomes
affected by
agency?

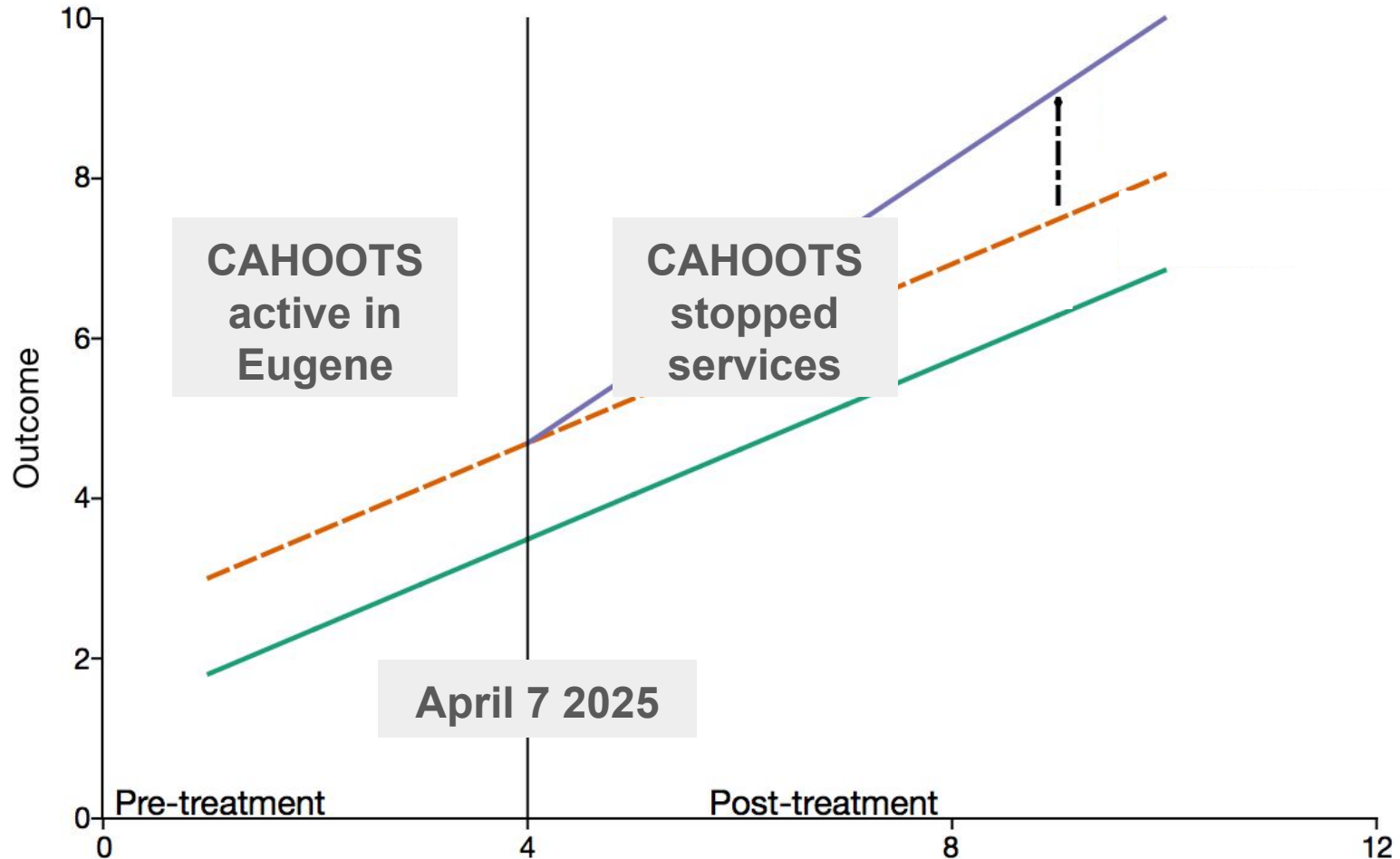
- Chi-squared
test
- Multinomial
logistic
regression

3

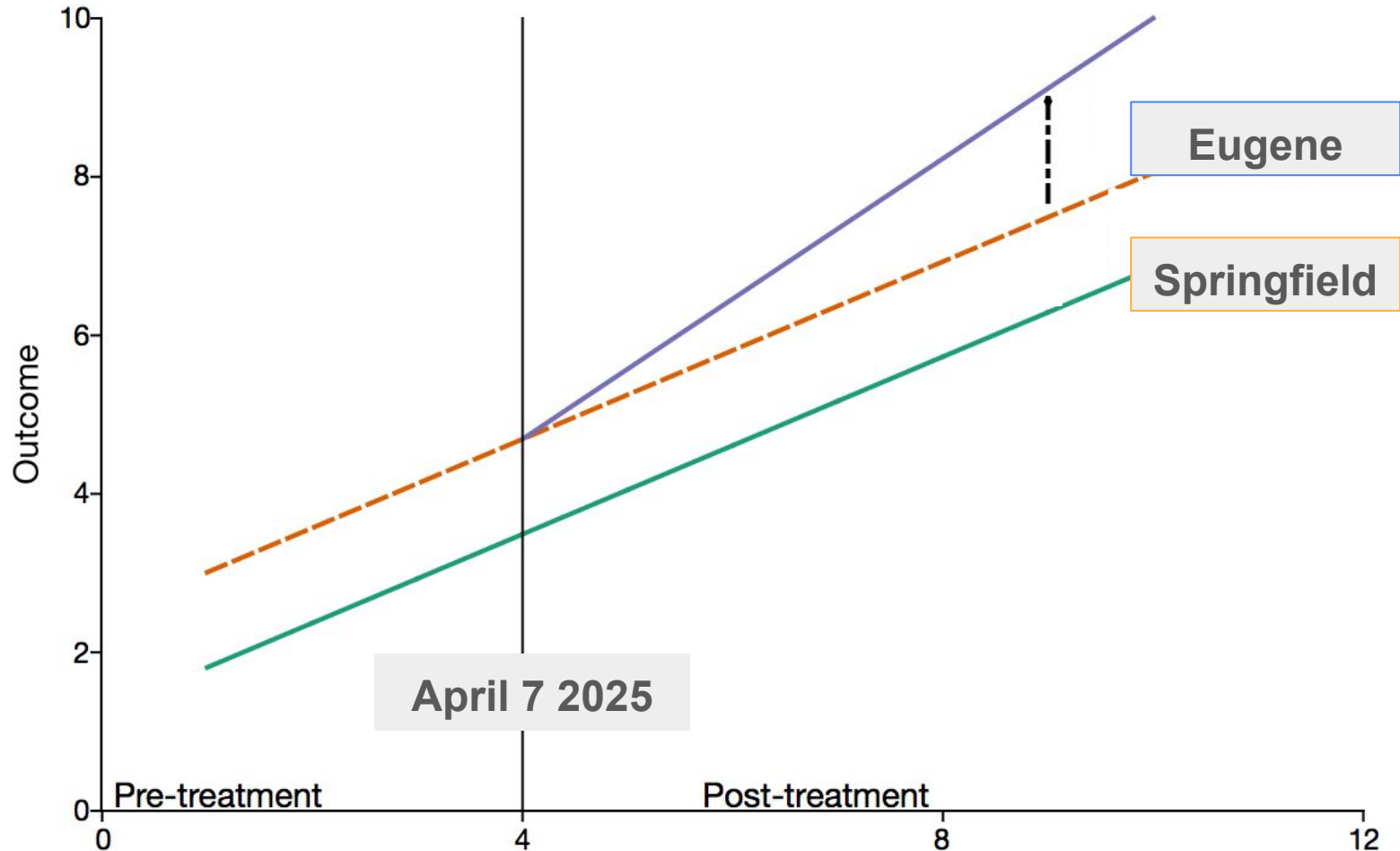
Is there a statistical
difference in call
outcomes due to
agency change?

-difference in
differences

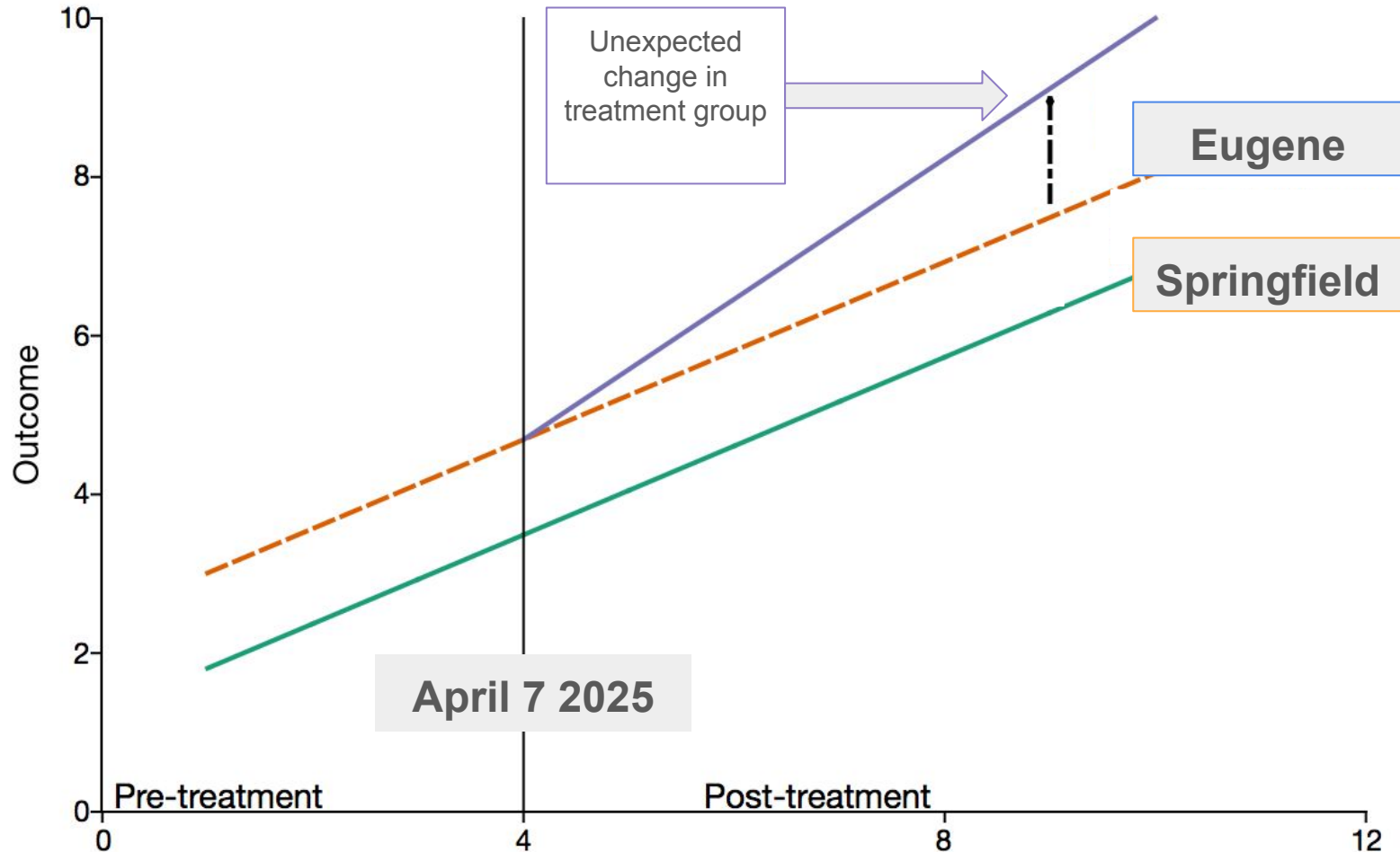
Difference-In-Differences Estimation



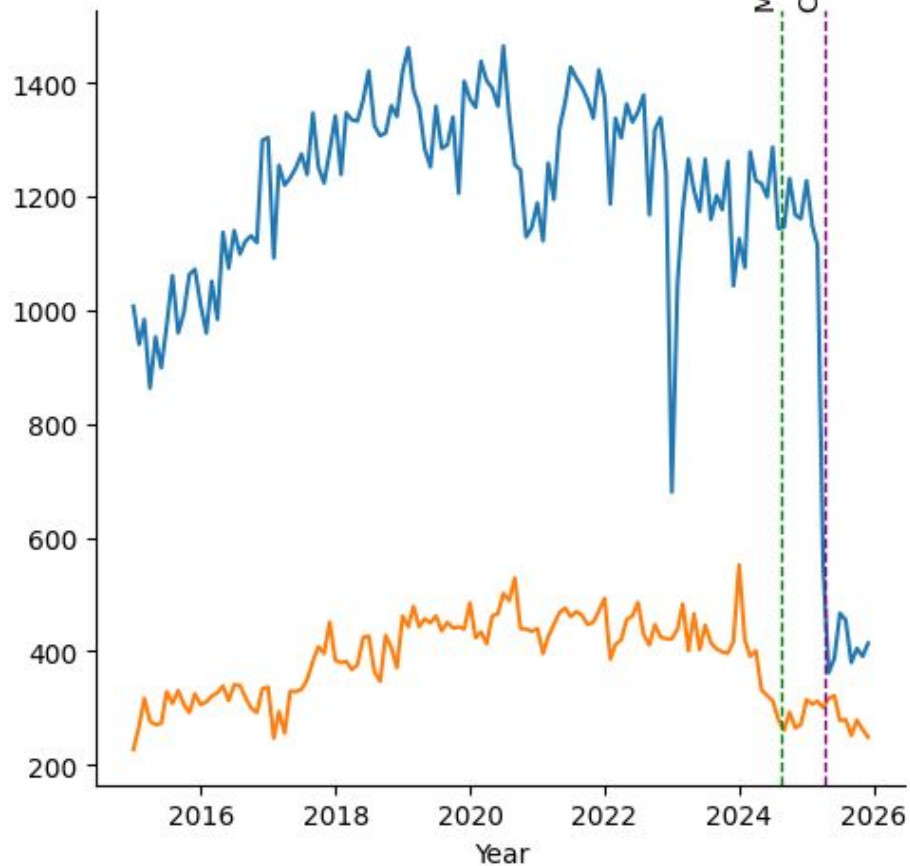
Difference-In-Differences Estimation



Difference-In-Differences Estimation

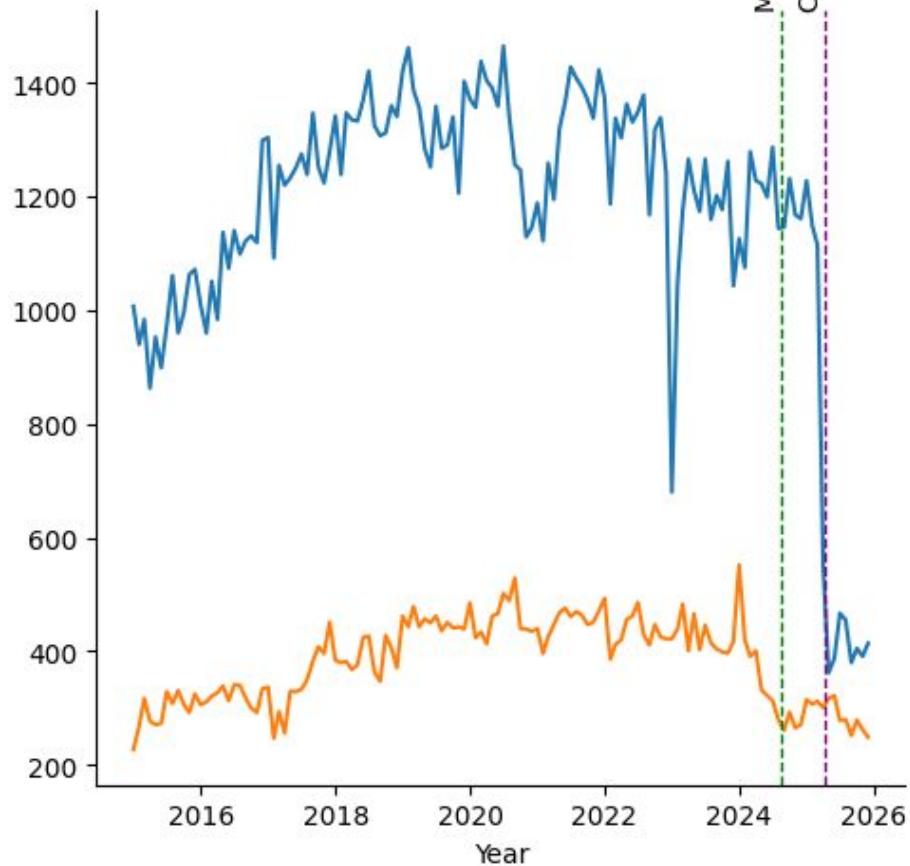


ASSISTED



Eugene
Springfield

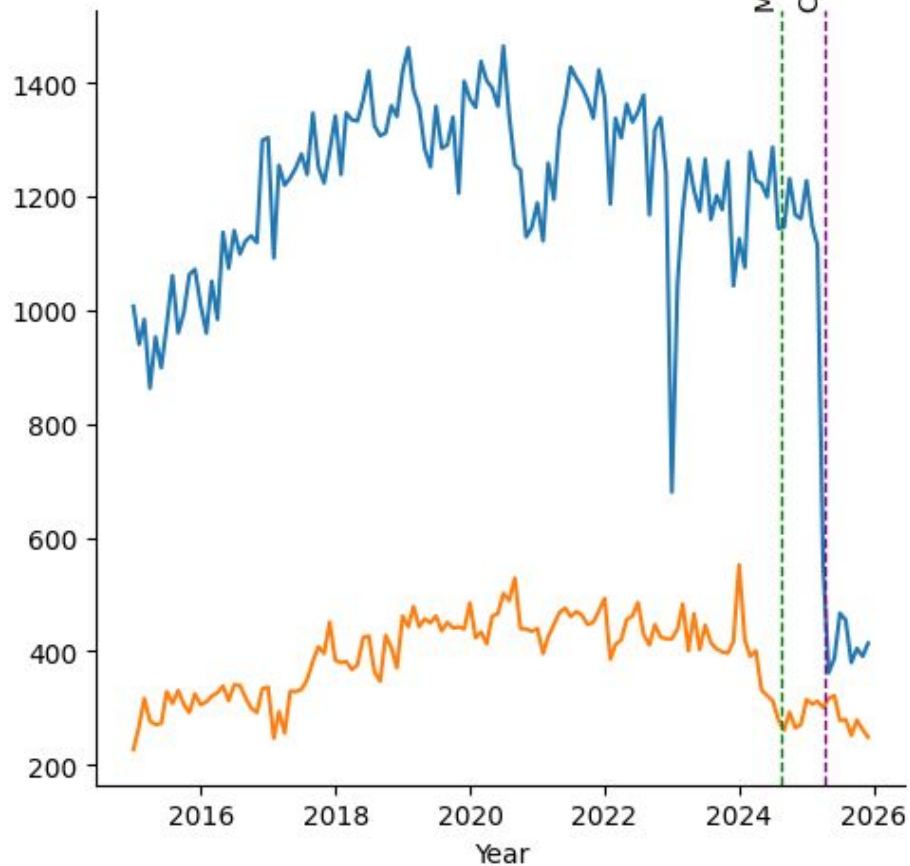
ASSISTED



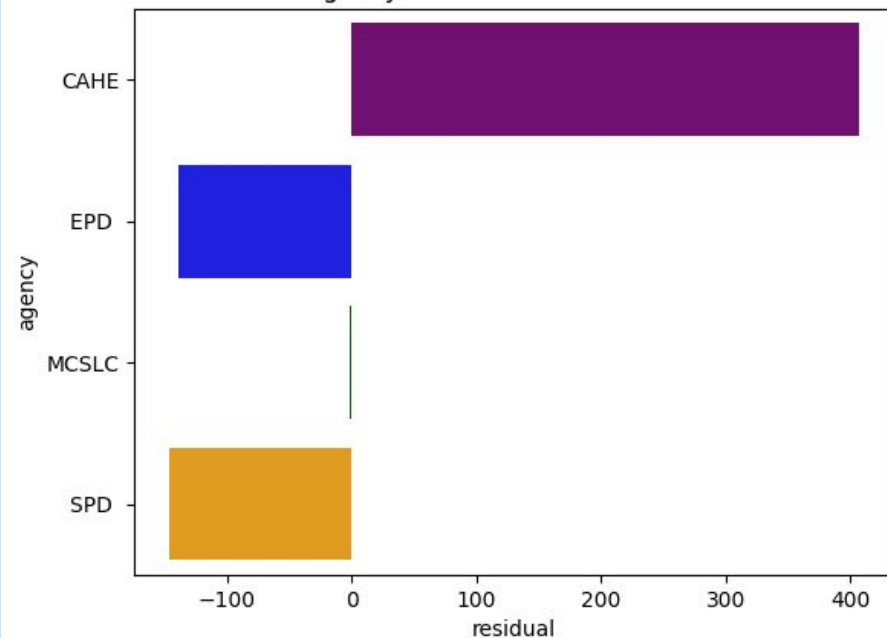
Eugene
Springfield

-7.924%

ASSISTED



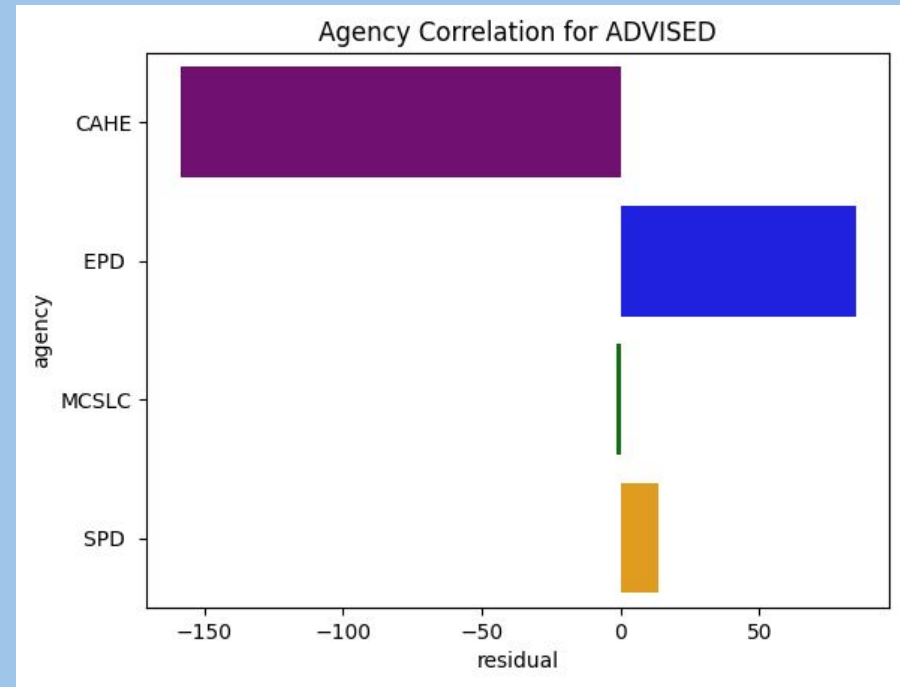
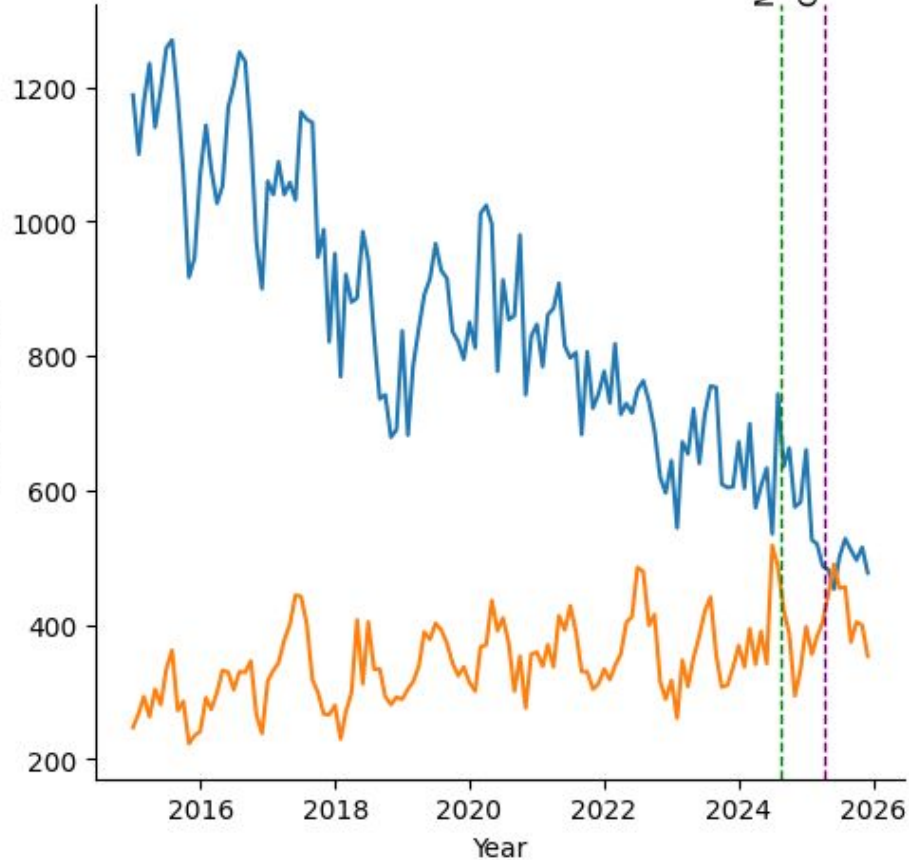
Agency Correlation for ASSISTED



Eugene
Springfield

-7.924%

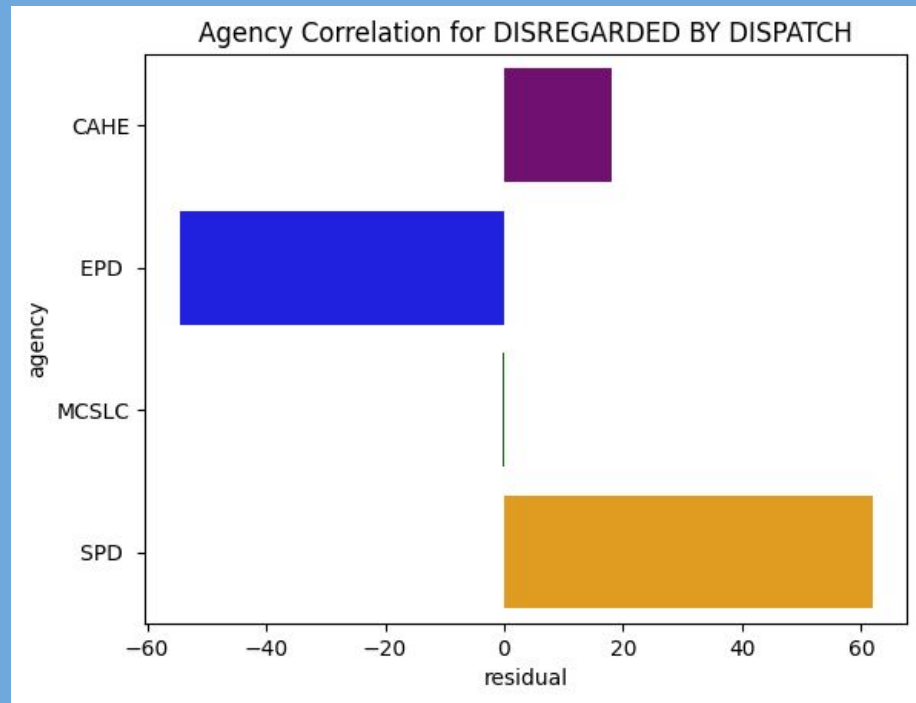
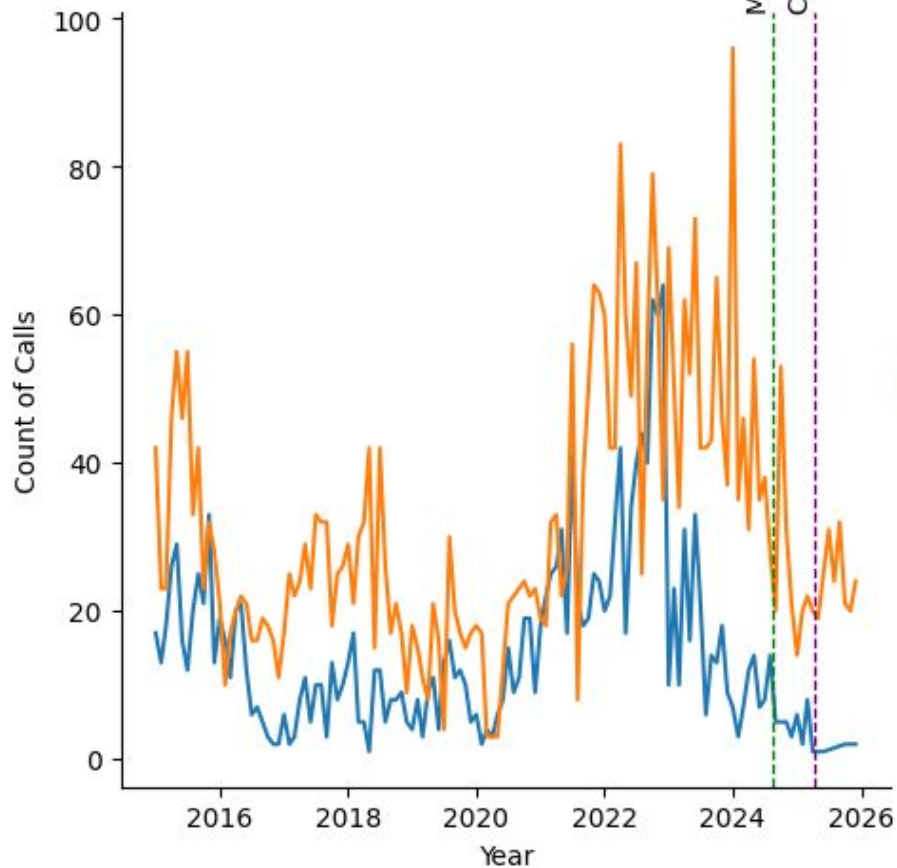
ADVISED



Eugene
Springfield

-7.102%

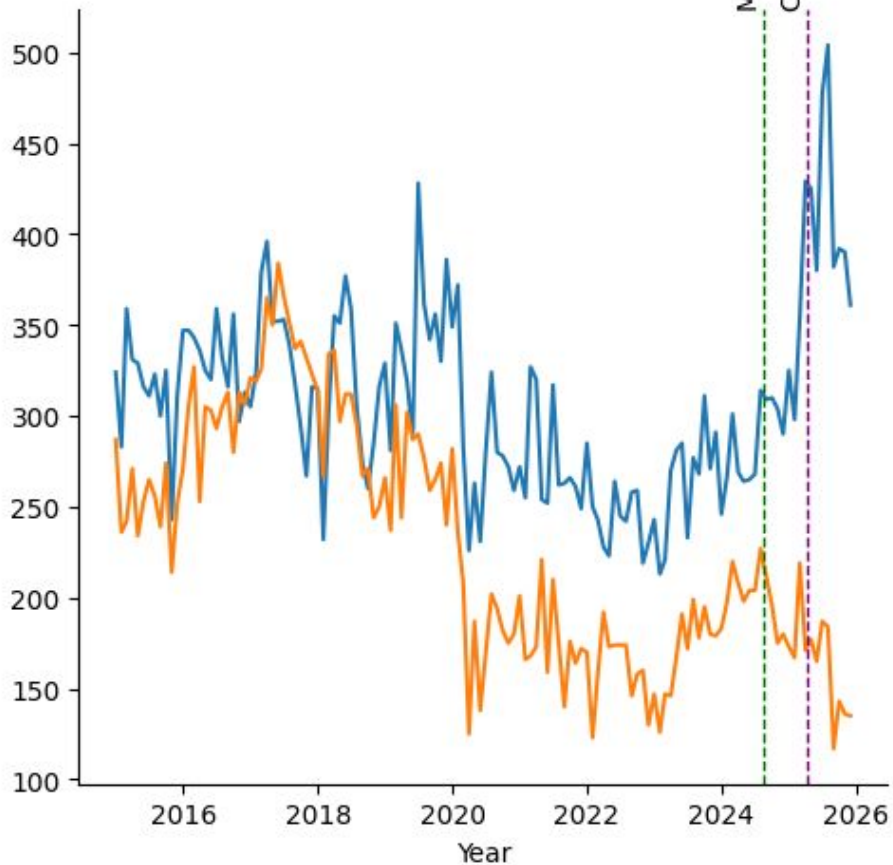
DISREGARDED BY DISPATCH



Eugene
Springfield

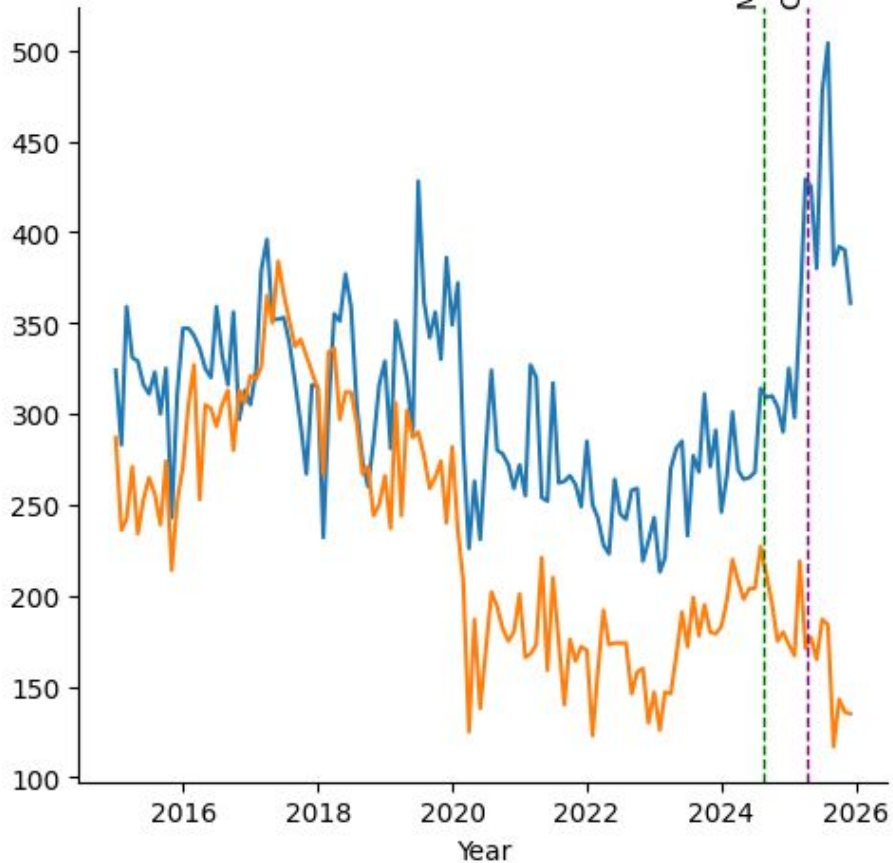
-0.098%

ARREST



Eugene
Springfield

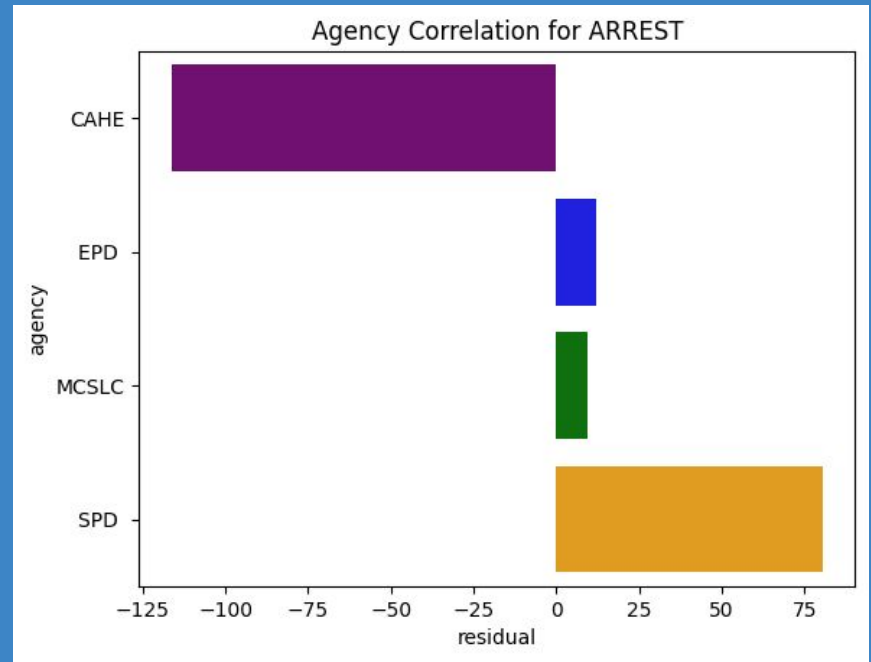
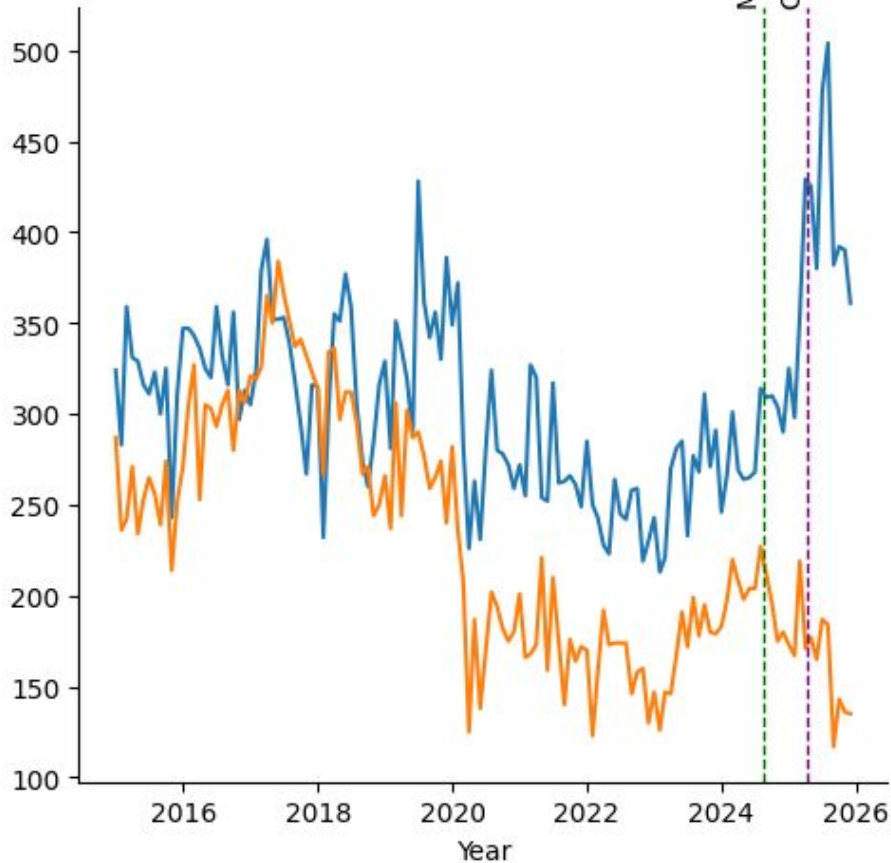
ARREST



Eugene
Springfield

3.2%

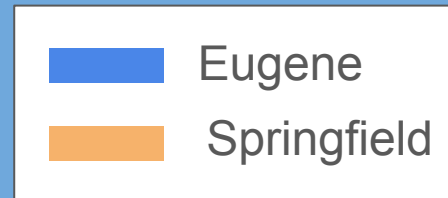
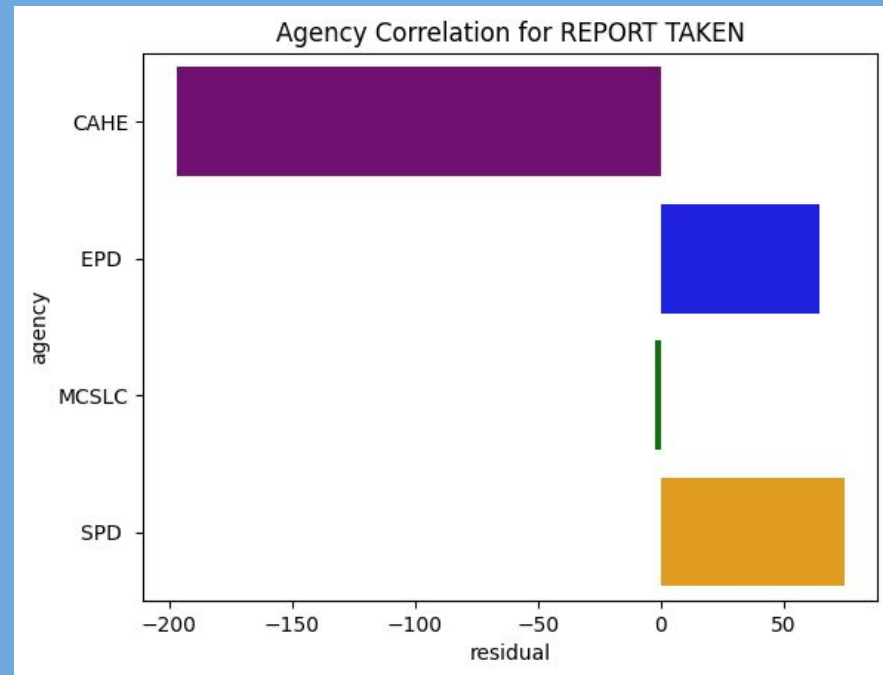
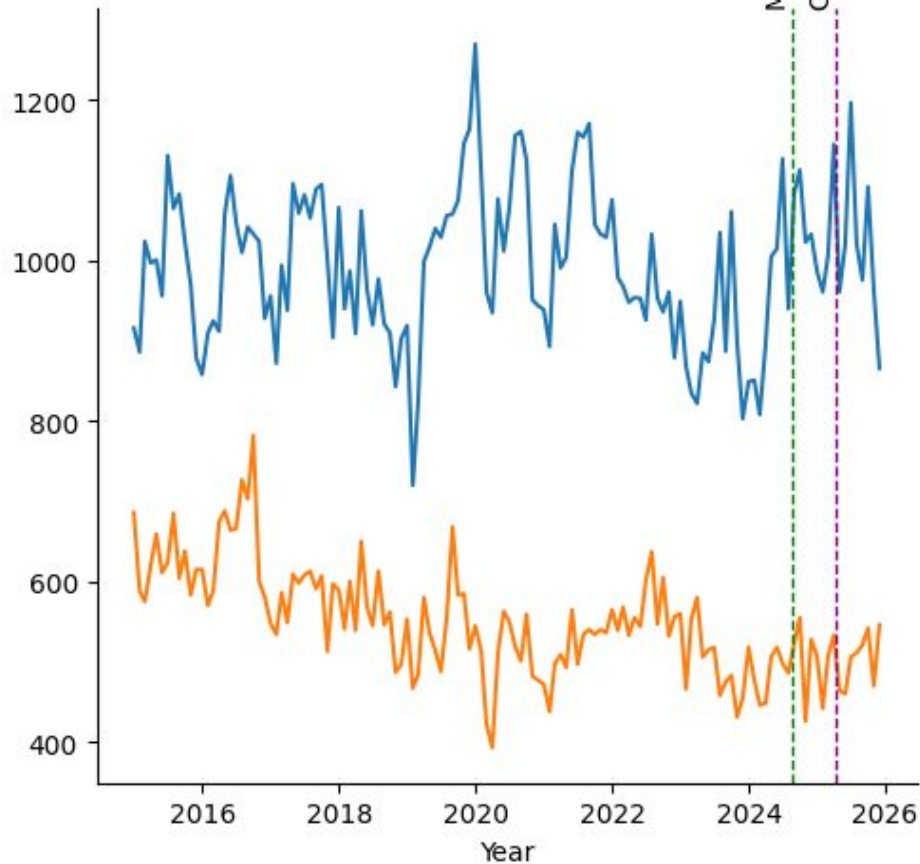
ARREST



Eugene
Springfield

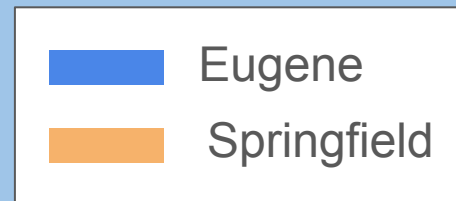
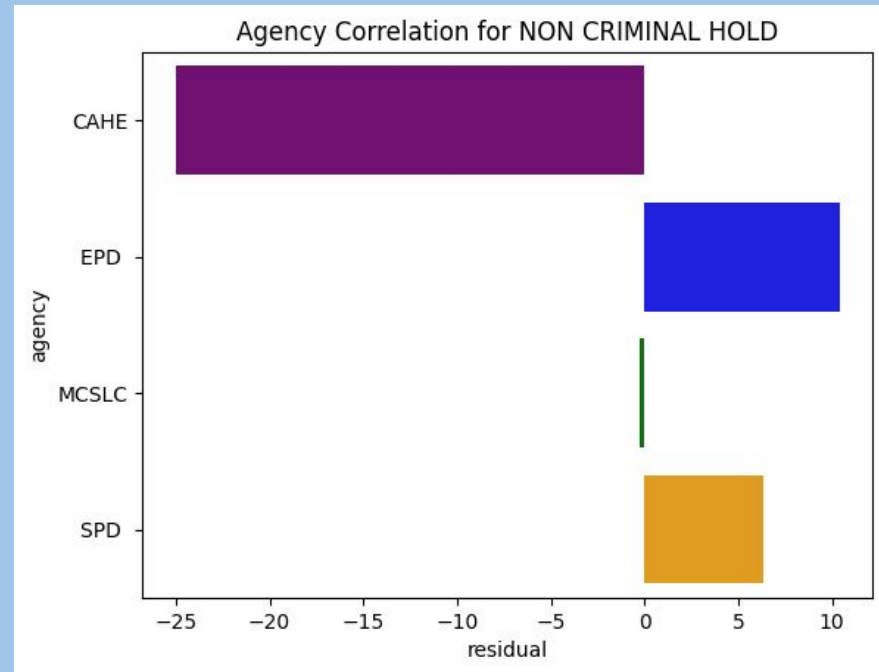
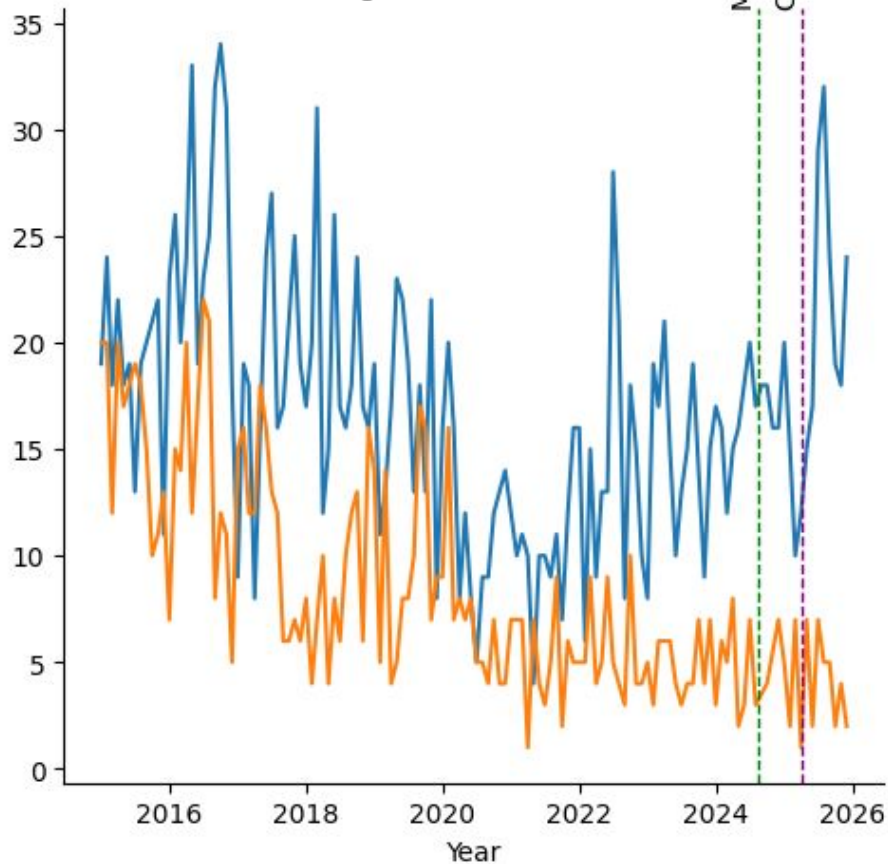
3.2%

REPORT TAKEN



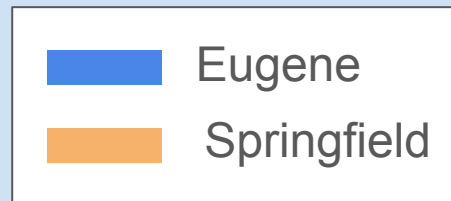
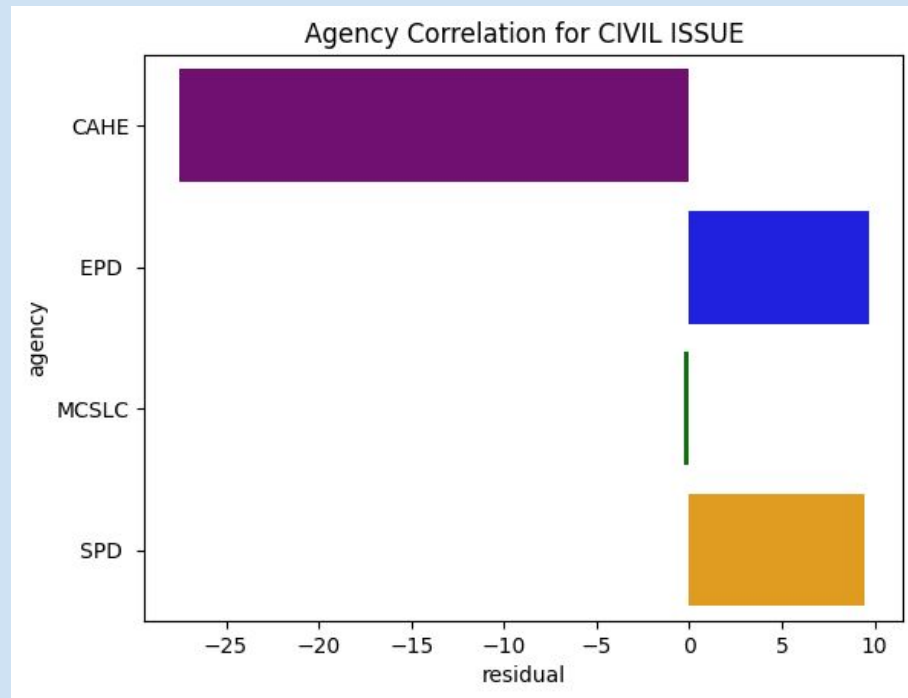
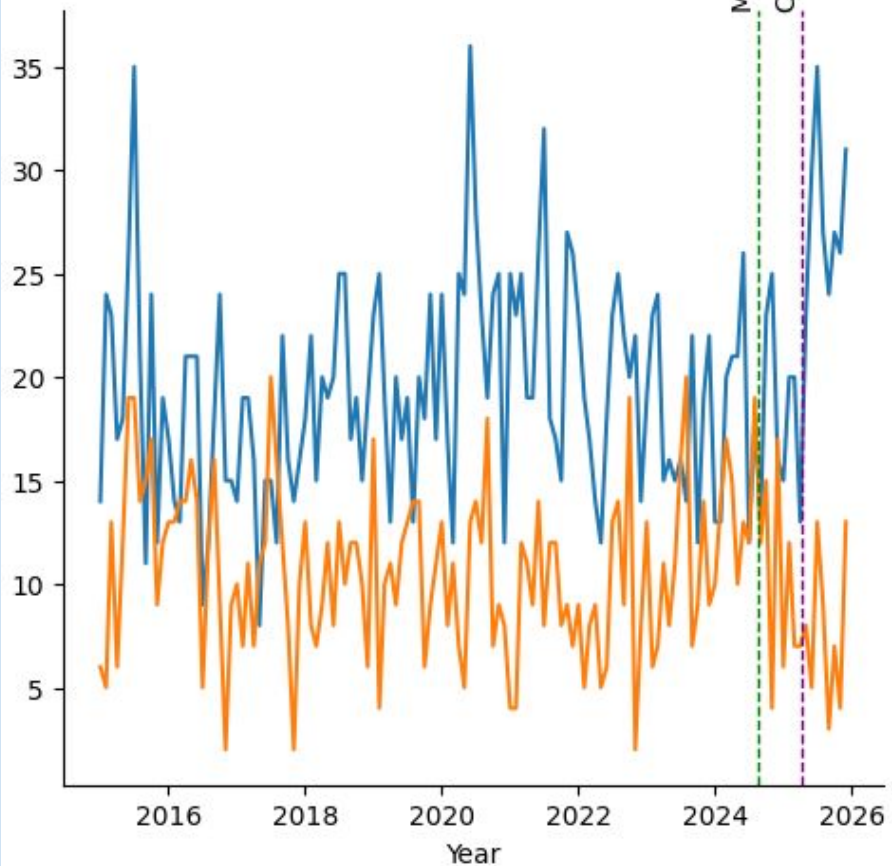
0.994%

NON CRIMINAL HOLD



0.185%

CIVIL ISSUE



0.166%

What does this all mean?

With everything considered it would be appropriate to conclude that there is a correlation between agency and call outcome. There are changes in call outcomes after CAHOOTS stopped services in Eugene

What does this all mean?

With appropriate changes, there could be a correlation between the two. There are changes in the data.



Acknowledgements

Thank you to Dr. Rori Rohlf for giving advice when I was stuck and for providing the data to be able to do this project.

Thank you to my classmates who have given me constructive feedback and made my project better from it.

Thank you to my boyfriend Travis for listening to me explain my project and practice presenting again and again.

QUESTIONS?